

# **COXIS-high tumor nests and LRRC15 CAF stroma form therapy-resistant niches in breast cancer**

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## **INTRODUCTION**

Breast cancer is a biologically and clinically heterogeneous disease in which therapeutic resistance remains a major cause of treatment failure. Although immune checkpoint blockade (ICB) has produced durable responses in some cancers, the majority of breast tumors do not respond, particularly those with immune-excluded or “cold” tumor microenvironments. Increasing evidence indicates that resistance is not determined solely by tumor cells, but emerges through interactions between malignant cells and surrounding stromal and immune components of the tumor microenvironment.

Cyclooxygenase-2 (COX-2) signaling and its associated COXIS transcriptional program have been implicated in inflammation-driven tumor progression. COX-2-derived prostaglandin E (PGE) promotes epithelial–mesenchymal transition (EMT), stress-adaptation programs, and suppression of anti-tumor immunity. In parallel, single-cell studies have identified a distinct subset of cancer-associated fibroblasts (CAFs) marked by LRRC15 expression. These LRRC15 CAFs arise in response to TGF- signaling, remodel extracellular matrix, and directly impair CD8 T-cell effector function, thereby limiting response to checkpoint blockade. Tumor cells may also adopt stress-response states such as the hypoxia-related drug-resistant (HRDS) program, enabling survival under therapy pressure.

However, most prior work has analyzed these processes using bulk transcriptomics or dissociated cells, which remove spatial architecture. The spatial arrangement of tumor cells, CAFs, and immune cells is increasingly recognized as critical for therapy response, yet the spatial relationship connecting COXIS-high tumor regions, LRRC15 CAF localization, EMT/HRDS programs, and ICB non-responder signatures is not well understood.

### **-Research objective**

This study used spatial transcriptomic data to evaluate how tumor-intrinsic COXIS signaling and microenvironmental resistance programs are organized within intact breast tumor tissue.

We hypothesized that within breast tumors, spatially localized COXIS-high tumor nests are adjacent to LRRC15 CAF-rich stroma, indicating spatially organized tumor–stroma ecosystems associated with therapy resistance.

The overall objective was to determine whether COX-2 signaling is spatially associated with stromal remodeling and immune therapy resistance in breast cancer.

## **METHODS AND MATERIALS**

### **-Study design**

This investigation was designed as a computational spatial transcriptomic analysis. Our study compared gene expression signatures and biological pathways in spatially defined tumor and stromal regions within breast cancer tissue sections.

### **-Data acquisition and preprocessing**

Spatial transcriptomic dataset (10x Genomix Visium) from breast tumors were analyzed. Raw count matrices and spatial coordinate information were imported as a spatial experiment.

Quality control steps included:

- removal of low-quality spots
- normalization of gene expression data
- Dimensionality reduction and Leiden clustering based on the normalized top 5% Highly Variable Genes(HVGs)

Spots were annotated into major tissue compartments using pathology annotations and Leiden clustering i.e. tumor regions and cancer-associated fibroblast (CAF)–rich stromal regions.

### **- Spatial signature scoring and group comparison**

To identify dominant transcriptional programs in an unbiased manner, we performed exploratory gene-signature scoring across all spatial spots without prior knowledge of which signatures would be relevant. Publicly available gene sets were systematically screened and scored, and enrichment patterns were evaluated across spatially defined groups. Signatures were not preselected based on expected biology but instead, they were interpreted only after enrichment patterns emerged from the data.

After scoring, CAF-rich and tumor-rich groups were compared statistically, and differences in signature enrichment between the groups were assessed using non-parametric tests.

### - Differential expression and pathway enrichment

Differentially expressed genes were identified between tumor-enriched spots vs CAF-enriched spots after which Gene Ontology (GO) and pathway enrichment analysis was used to identify biological processes preferentially activated in each region.

Adjusted p-values were calculated using false discovery rate (FDR) correction. Statistical significance was defined as **adjusted p < 0.05**.

Figures were generated to display pathway enrichment and signature distribution among the Tumor and CAF groups.

## DATA LOADING AND EXPLORATION

```
library(scater)
```

```
Loading required package: SingleCellExperiment
```

```
Loading required package: SummarizedExperiment
```

```
Loading required package: MatrixGenerics
```

```
Loading required package: matrixStats
```

```
Attaching package: 'MatrixGenerics'
```

```
The following objects are masked from 'package:matrixStats':
```

```
colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
```

```
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,  
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,  
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,  
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,  
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,  
rowWeightedSds, rowWeightedVars
```

Loading required package: GenomicRanges

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Loading required package: S4Vectors

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomeInfoDb

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with  
'browseVignettes()'. To cite Bioconductor, see  
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

Loading required package: scuttle

Loading required package: ggplot2

Warning: package 'ggplot2' was built under R version 4.5.1

```
library(scran)
library(scuttle)
library(SpatialExperiment)
library(ggspavis)
library(ComplexHeatmap)
```

Warning: package 'ComplexHeatmap' was built under R version 4.5.1

Loading required package: grid

=====

ComplexHeatmap version 2.24.1

Bioconductor page: <http://bioconductor.org/packages/ComplexHeatmap/>

Github page: <https://github.com/jokergoo/ComplexHeatmap>

Documentation: <http://jokergoo.github.io/ComplexHeatmap-reference>

If you use it in published research, please cite either one:

- Gu, Z. Complex Heatmap Visualization. iMeta 2022.
- Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional genomic data. Bioinformatics 2016.

The new InteractiveComplexHeatmap package can directly export static complex heatmaps into an interactive Shiny app with zero effort. Have a try!

This message can be suppressed by:

```
suppressPackageStartupMessages(library(ComplexHeatmap))
```

=====

```
library(igraph)
```

Attaching package: 'igraph'

The following object is masked from 'package:scater':

normalize

The following object is masked from 'package:GenomicRanges':

union

The following object is masked from 'package:IRanges':

union

The following object is masked from 'package:S4Vectors':

union

The following objects are masked from 'package:BiocGenerics':

normalize, path, union

The following objects are masked from 'package:generics':

components, union

The following objects are masked from 'package:stats':

decompose, spectrum

The following object is masked from 'package:base':

union

```
library(patchwork)
```

Warning: package 'patchwork' was built under R version 4.5.1

```
library(ggplot2)
library(pals)
```

```
Filter(function(x) is(get(x), "SpatialExperiment"), ls())
```

```
character(0)
```

```
data = load(file = "Breast_ST6.RData")
```

```
speBreast
```

```
class: SpatialExperiment
dim: 17943 355
metadata(0):
assays(1): counts
rownames(17943): ENSG000000187634 ENSG000000188976 ... ENSG000000154620
               ENSG00000012817
rowData names(1): symbol
colnames(355): AAACGAGACGGTTGAT-1 AAAGTGTGATTTATCT-1 ...
               TTGACTATTGTCCGGC-1 TTGCCAAGCAGAACCC-1
colData names(5): in_tissue array_row array_col sample_id annotation
reducedDimNames(0):
mainExpName: NULL
altExpNames(0):
spatialCoords names(2) : pxl_col_in_fullres pxl_row_in_fullres
imgData names(4): sample_id image_id data scaleFactor
```

```
dim(speBreast)
```

```
[1] 17943    355
```

```
assayNames(speBreast)
```

```
[1] "counts"
```

```
head(rowData(speBreast))
```

```
DataFrame with 6 rows and 1 column
      symbol
<character>
ENSG000000187634 SAMD11
ENSG000000188976 NOC2L
ENSG000000187961 KLHL17
ENSG000000187583 PLEKHN1
ENSG000000187642 PERM1
ENSG000000188290 HES4
```



```
head(colData(speBreast))
```

DataFrame with 6 rows and 5 columns

|                    | in_tissue | array_row | array_col | sample_id   | annotation  |
|--------------------|-----------|-----------|-----------|-------------|-------------|
|                    | <logical> | <integer> | <integer> | <character> | <character> |
| AAACGAGACGGTTGAT-1 | TRUE      | 35        | 79        | sample01    | stroma      |
| AAAGTGTGATTTATCT-1 | TRUE      | 44        | 94        | sample01    | stroma      |
| AAAGTTGACTCCCGTA-1 | TRUE      | 42        | 96        | sample01    | stroma      |
| AACCCTACTGTCAATA-1 | TRUE      | 40        | 96        | sample01    | CAF         |
| AACCGCTAAGGGATGC-1 | TRUE      | 46        | 92        | sample01    | stroma      |
| AACTACCCGTTTGTCA-1 | TRUE      | 47        | 113       | sample01    | necrosis    |

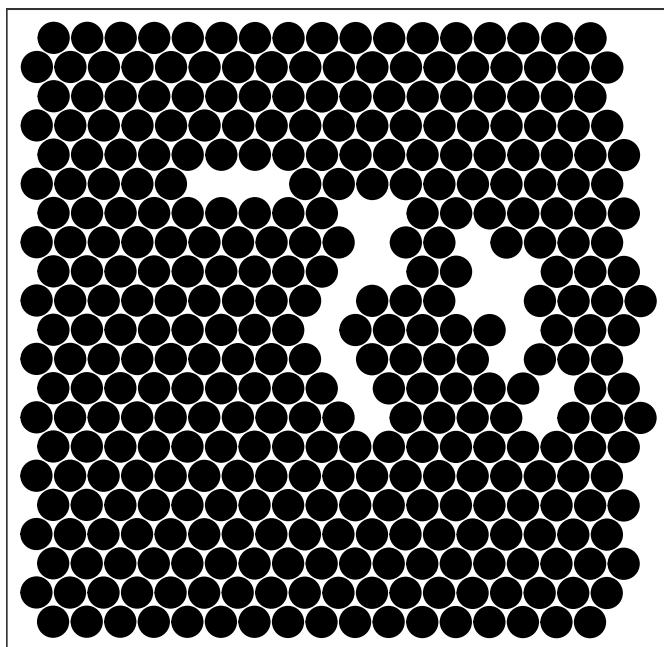
```
head(spatialCoords(speBreast))
```

|                    | pxl_col_in_fullres | pxl_row_in_fullres |
|--------------------|--------------------|--------------------|
| AAACGAGACGGTTGAT-1 | 16601              | 12477              |
| AAAGTGTGATTTATCT-1 | 18775              | 14750              |
| AAAGTTGACTCCCGTA-1 | 19066              | 14246              |
| AACCCTACTGTCAATA-1 | 19066              | 13741              |
| AACCGCTAAGGGATGC-1 | 18485              | 15255              |
| AACTACCCGTTTGTCA-1 | 21530              | 15510              |

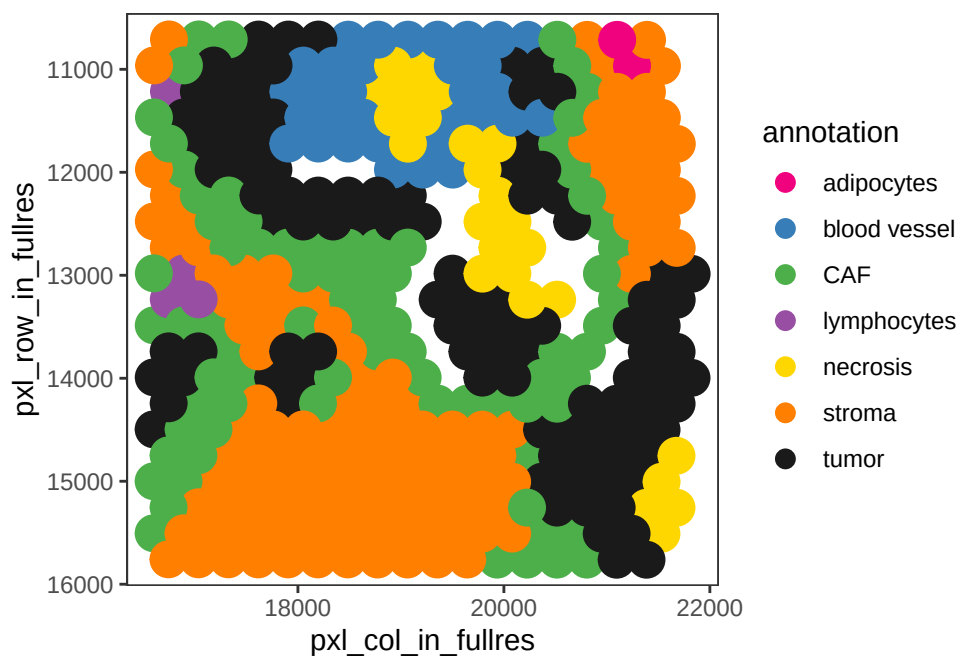
## DATA VISUALIZATION (ggpavis)

```
# plot spatial coordinates (spots)
plotCoords(speBreast, point_size = 5,)
```

Warning: `aes\_string()` was deprecated in ggplot2 3.0.0.  
i Please use tidy evaluation idioms with `aes()`.  
i See also `vignette("ggplot2-in-packages")` for more information.  
i The deprecated feature was likely used in the ggspavis package.  
Please report the issue at <<https://github.com/lmweber/ggspavis/issues>>.



```
# plot spot by annotation
plotCoords(speBreast, annotate = "annotation",
           point_size = 6, pal = "libd_layer_colors", show_axes = TRUE)
```



## Top-left spot annotation

```
coordinates = spatialCoords(speBreast)
colnames(coordinates) = c("x","y")
coordinates = as.data.frame(coordinates)

head(coordinates)
```

|                    | x     | y     |
|--------------------|-------|-------|
| AAACGAGACGGTTGAT-1 | 16601 | 12477 |
| AAAGTGTGATTTATCT-1 | 18775 | 14750 |
| AAAGTTGACTCCCGTA-1 | 19066 | 14246 |
| AACCCTACTGTCAATA-1 | 19066 | 13741 |
| AACCGCTAAGGGATGC-1 | 18485 | 15255 |
| AACTACCCGTTTGTCA-1 | 21530 | 15510 |

```
range(coordinates$x)
```

```
[1] 16599 21823
```

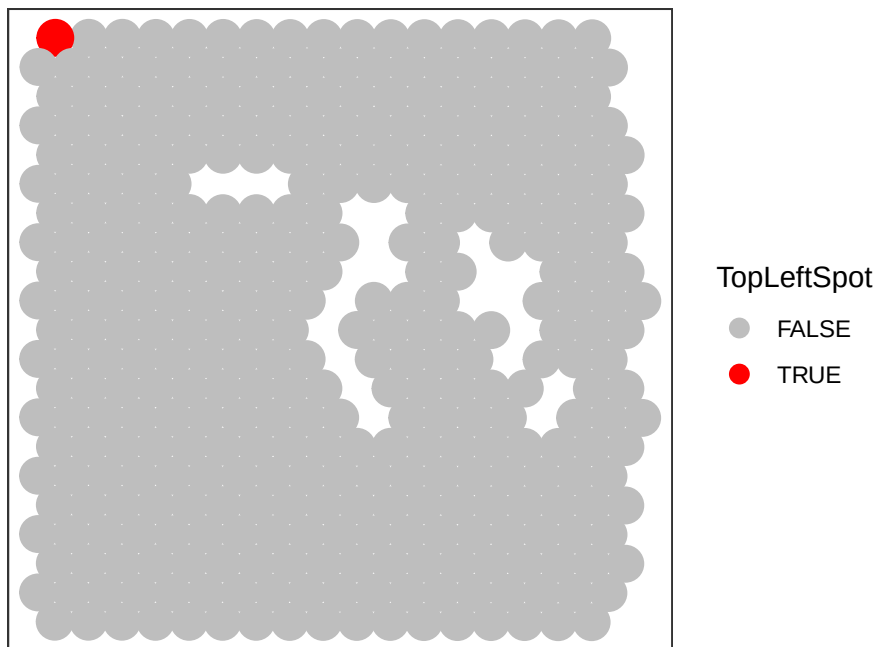
```
range(coordinates$y)
```

```
[1] 10710 15762
```

```
## top-left spot red
Min_y_Coord <- coordinates$y == min(coordinates$y)
Min_x_Coord <- coordinates$x == min(coordinates$x[Min_y_Coord])

TopLeftSpot <- rownames(coordinates)[which(Min_x_Coord & Min_y_Coord)]
colData(speBreast)[, "TopLeftSpot"] <- as.character(colnames(speBreast)==TopLeftSpot)

plotCoords(speBreast, annotate = "TopLeftSpot", point_size = 6, pal = c("grey", "red"))
```



## QUALITY CONTROL (scater library)

```
speBreast <- addPerCellQC(speBreast)
head(colData(speBreast))
```

DataFrame with 6 rows and 9 columns

|                    | in_tissue   | array_row | array_col | sample_id   | annotation  |
|--------------------|-------------|-----------|-----------|-------------|-------------|
|                    | <logical>   | <integer> | <integer> | <character> | <character> |
| AAACGAGACGGTTGAT-1 | TRUE        | 35        | 79        | sample01    | stroma      |
| AAAGTGTGATTTATCT-1 | TRUE        | 44        | 94        | sample01    | stroma      |
| AAAGTTGACTCCCGTA-1 | TRUE        | 42        | 96        | sample01    | stroma      |
| AACCCTACTGTCAATA-1 | TRUE        | 40        | 96        | sample01    | CAF         |
| AACCGCTAAGGGATGC-1 | TRUE        | 46        | 92        | sample01    | stroma      |
| AACTACCCGTTTGTCA-1 | TRUE        | 47        | 113       | sample01    | necrosis    |
|                    | TopLeftSpot | sum       | detected  | total       |             |
|                    | <character> | <numeric> | <integer> | <numeric>   |             |
| AAACGAGACGGTTGAT-1 | FALSE       | 11825     | 4856      | 11825       |             |
| AAAGTGTGATTTATCT-1 | FALSE       | 8121      | 3599      | 8121        |             |
| AAAGTTGACTCCCGTA-1 | FALSE       | 9967      | 4491      | 9967        |             |
| AACCCTACTGTCAATA-1 | FALSE       | 13141     | 5351      | 13141       |             |
| AACCGCTAAGGGATGC-1 | FALSE       | 9794      | 4079      | 9794        |             |

AACTACCCGTTTGTCA-1

FALSE

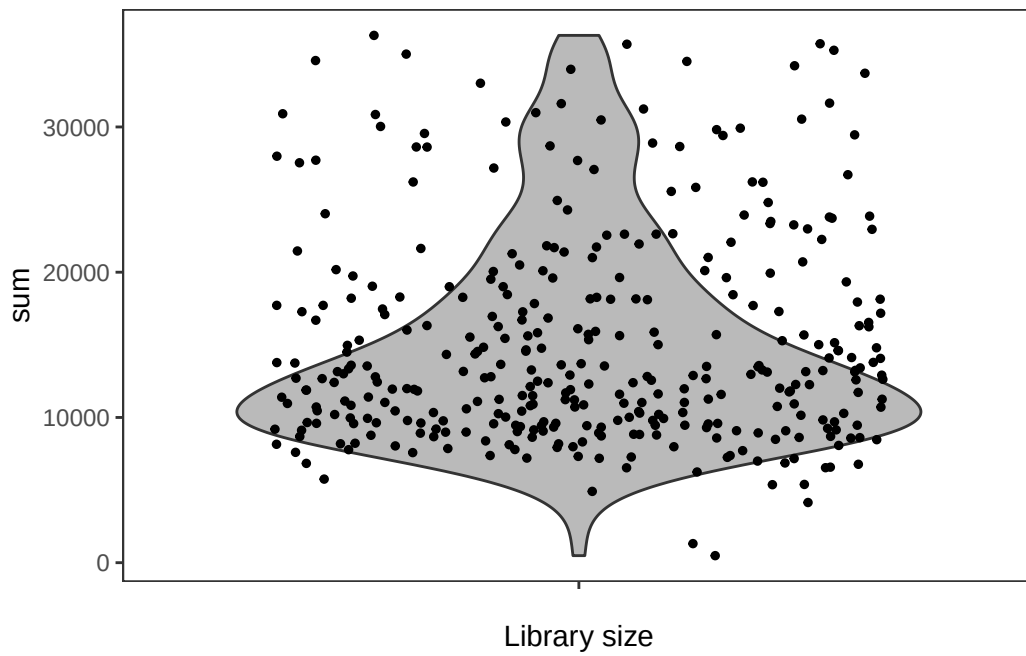
31607

7247

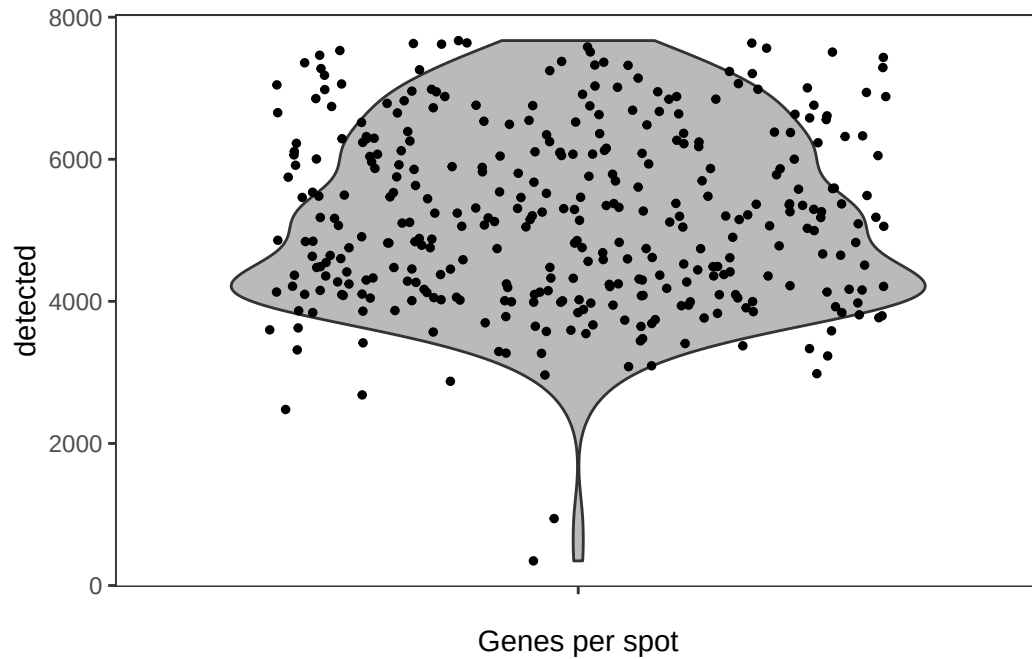
31607

## QC metrics Visualization

```
plotObsQC(speBreast, plot_type = "violin", x_metric = "sum", point_size = 1) +  
  xlab("Library size")
```

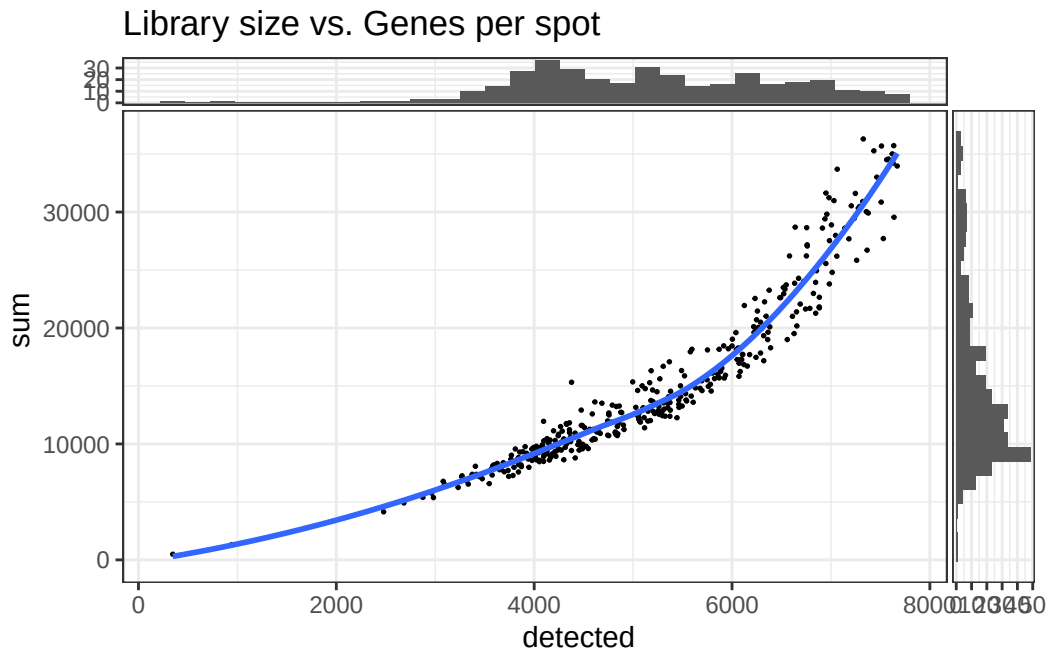


```
plotObsQC(speBreast, plot_type = "violin", x_metric = "detected", point_size = 1) +  
  xlab("Genes per spot")
```



```
plotObsQC(
  speBreast, plot_type = "scatter", x_metric = "detected", y_metric = "sum") +
  ggtitle("Library size vs. Genes per spot") + labs(x = "detected", y = "sum")
```

```
`geom_smooth()` using formula = 'y ~ x'
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```

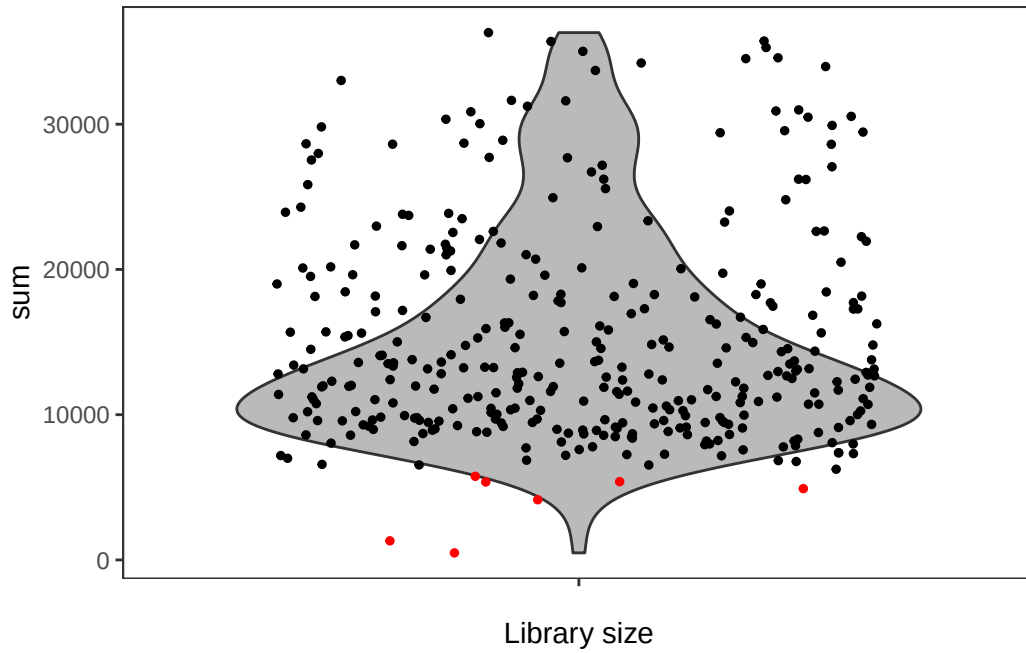


Threshold selection.

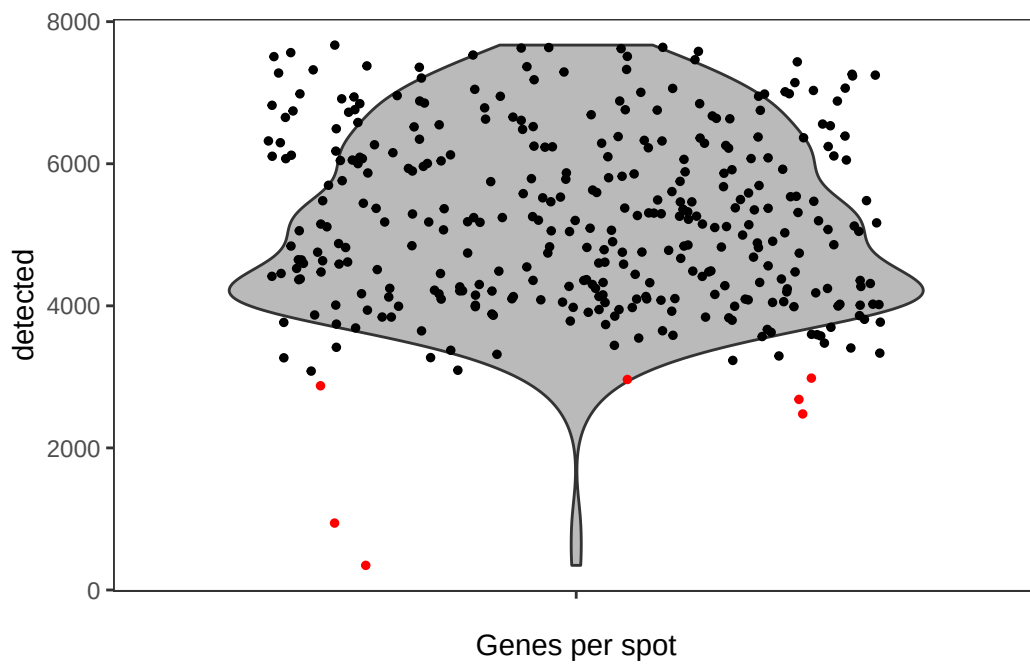
```
speBreast$qc_lib_size <- speBreast$sum < 6000
speBreast$qc_detected <- speBreast$detected < 3000
table(speBreast$qc_detected, speBreast$qc_lib_size)
```

|       | FALSE | TRUE |
|-------|-------|------|
| FALSE | 348   | 0    |
| TRUE  | 0     | 7    |

```
plotObsQC(speBreast, plot_type = "violin", x_metric = "sum",
          annotate = "qc_lib_size", point_size = 1) + xlab("Library size")
```



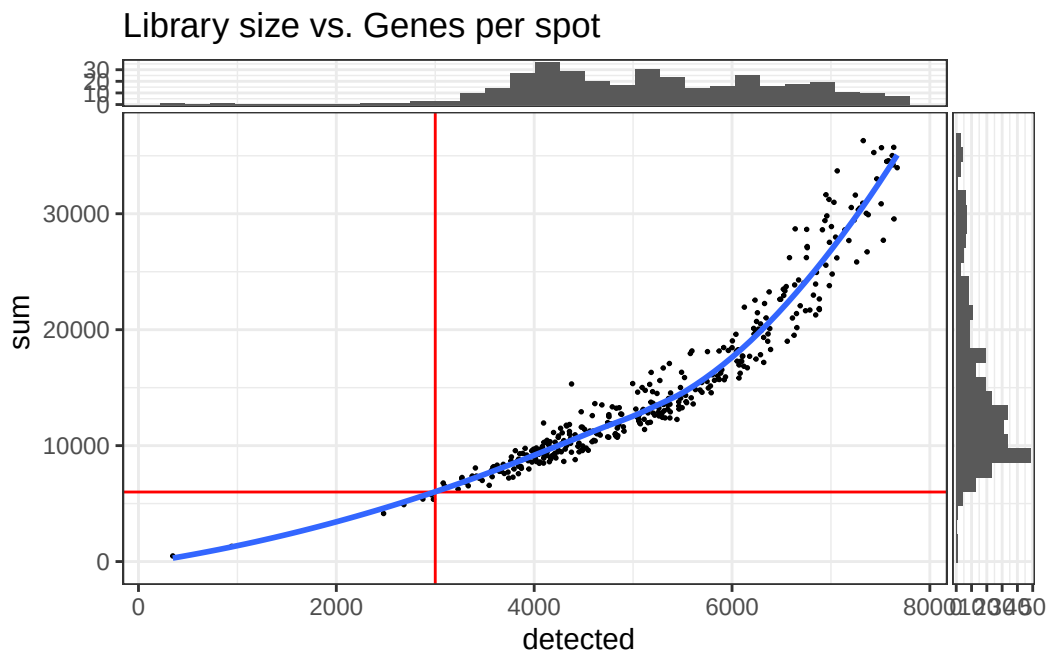
```
plotObsQC(speBreast, plot_type = "violin", x_metric = "detected",
  annotate = "qc_detected", point_size = 1) + xlab("Genes per spot")
```





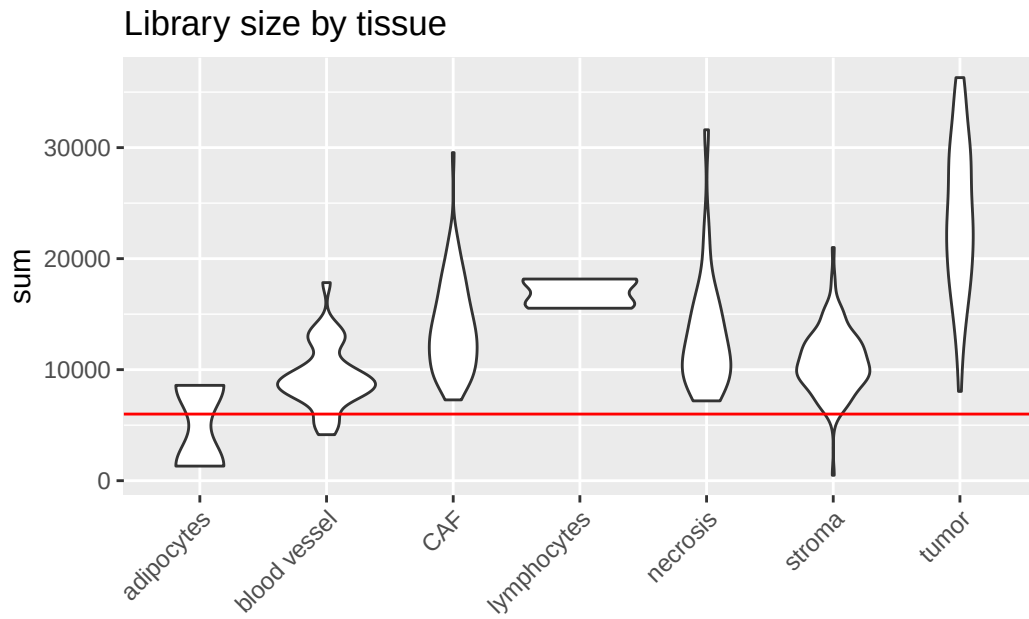
```
plotObsQC(
  speBreast, plot_type = "scatter", x_metric = "detected",
  y_metric = "sum", x_threshold = 3000, y_threshold = 6000) +
  ggtitle("Library size vs. Genes per spot") + labs(x = "detected", y = "sum")
```

```
`geom_smooth()` using formula = 'y ~ x'
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```

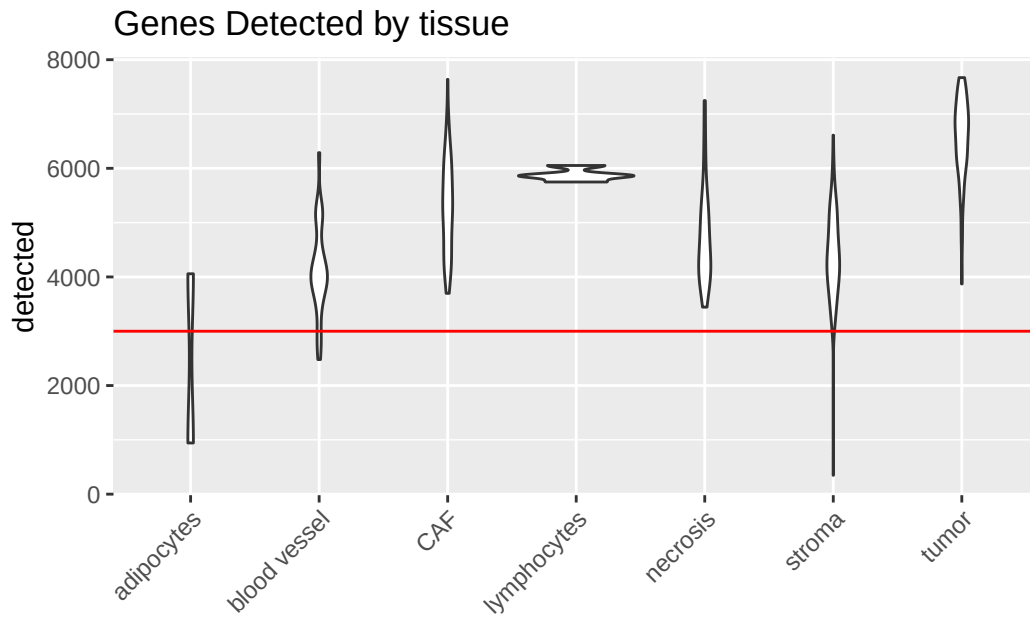


QC metrics visualization by different pathology annotations.

```
ggplot() + geom_violin(mapping = aes(x = speBreast$annotation, y = speBreast$sum)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ggtitle("Library size by tissue") + xlab("") + ylab("sum") +
  geom_hline(aes(yintercept = 6000), colour = "red")
```



```
ggplot() + geom_violin(mapping = aes(x = speBreast$annotation, y = speBreast$detected)) +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  
  ggtitle("Genes Detected by tissue") + xlab("") + ylab("detected") +  
  geom_hline(aes(yintercept = 3000), colour = "red")
```



As observed, Adipocytes have lowest UMIs and genes detected and this could be because they are mostly fat thus containing very little RNA.

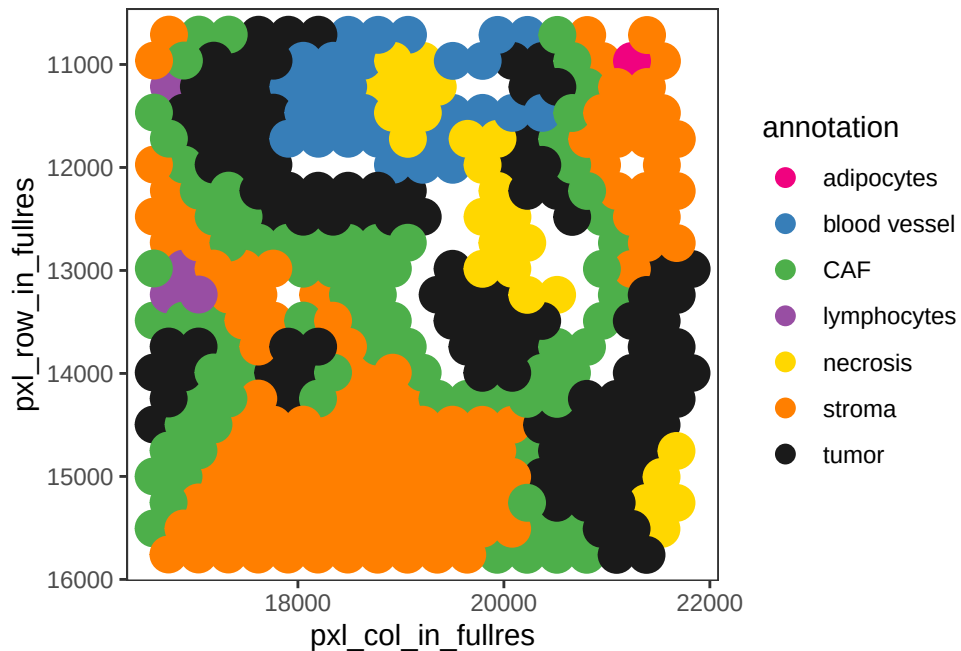
### Low-quality spots removal

```
speBreast$outliers <- speBreast$qc_lib_size | speBreast$qc_detected
table(speBreast$outliers, speBreast$annotation)
```

|       | adipocytes | blood vessel | CAF | lymphocytes | necrosis | stroma | tumor |
|-------|------------|--------------|-----|-------------|----------|--------|-------|
| FALSE | 1          | 28           | 77  | 4           | 25       | 113    | 100   |
| TRUE  | 1          | 4            | 0   | 0           | 0        | 2      | 0     |

```
speBreast <- speBreast[, !speBreast$outliers]
```

```
plotCoords(speBreast, annotate = "annotation",
            point_size = 6, pal = "libd_layer_colors", show_axes = TRUE)
```



## NORMALIZATION (scrn)

Getting rid of all the genes whose expression is zero in all the spots.

```
dim(speBreast)
```

```
[1] 17943  348
```

```
speBreast <- speBreast[apply(assay(speBreast) > 0, 1, any),]
```

```
dim(speBreast)
```

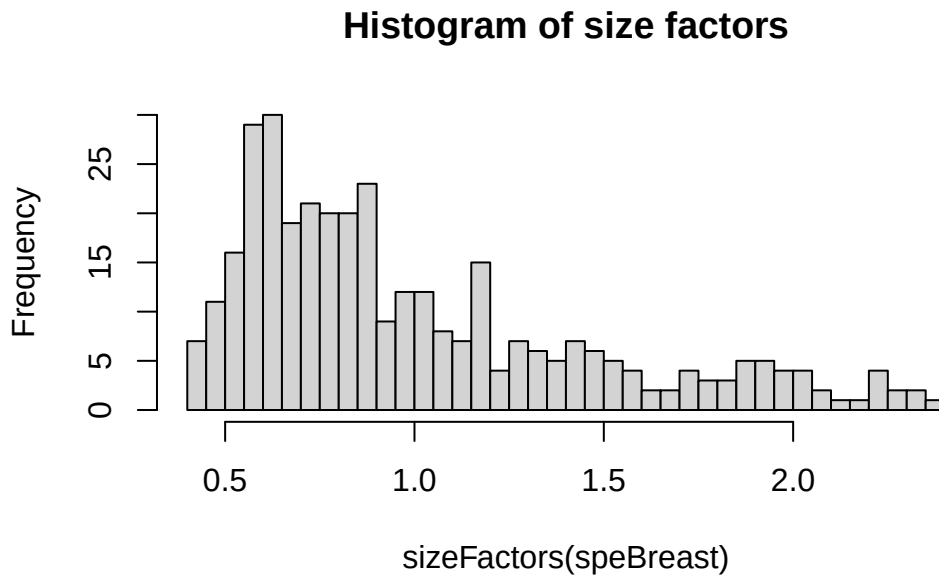
```
[1] 15898  348
```

Then, calculate library size factors and normalize data.

```
# calculate library size factors
speBreast <- computeLibraryFactors(speBreast)
summary(sizeFactors(speBreast))
```

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.4060 | 0.6368  | 0.8539 | 1.0000 | 1.2356  | 2.3608 |

```
hist(sizeFactors(speBreast), breaks = 50, main = "Histogram of size factors")
```



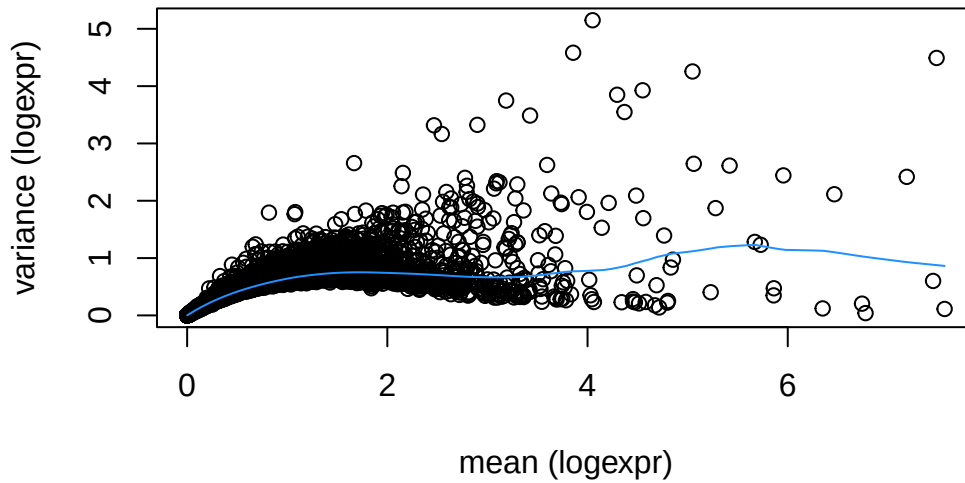
```
# calculate logcounts and store the output in a new assay
speBreast <- logNormCounts(speBreast)
assayNames(speBreast)
```

```
[1] "counts"    "logcounts"
```

## Feature selection based on top Highly Variable Genes(HVGs)

```
# fit mean-variance relationship
dec <- modelGeneVar(speBreast)

# visualize mean-variance relationship
plot(dec$mean, dec$total, xlab = "mean (logexpr)", ylab = "variance (logexpr)")
curve(metadata(dec)$trend(x), add = TRUE, col = "dodgerblue")
```



```
dec
```

```
DataFrame with 15898 rows and 6 columns
```

|                 | mean      | total     | tech      | bio         | p.value   | FDR       |
|-----------------|-----------|-----------|-----------|-------------|-----------|-----------|
|                 | <numeric> | <numeric> | <numeric> | <numeric>   | <numeric> | <numeric> |
| ENSG00000187634 | 0.0552722 | 0.0624187 | 0.0603078 | 0.00211088  | 0.476748  | 0.661672  |
| ENSG00000188976 | 0.5163318 | 0.3955543 | 0.4239155 | -0.02836120 | 0.544378  | 0.661672  |
| ENSG00000187961 | 0.1152458 | 0.1198071 | 0.1209107 | -0.00110359 | 0.506067  | 0.661672  |
| ENSG00000187583 | 0.0839241 | 0.0803361 | 0.0898814 | -0.00954534 | 0.570223  | 0.661672  |
| ENSG00000188290 | 0.7702380 | 0.6061659 | 0.5524259 | 0.05373992  | 0.435621  | 0.661672  |
| ...             | ...       | ...       | ...       | ...         | ...       | ...       |
| ENSG00000155961 | 0.0278028 | 0.026687  | 0.0308923 | -0.00420525 | 0.589712  | 0.661672  |
| ENSG00000155962 | 0.3101254 | 0.337756  | 0.2871967 | 0.05055956  | 0.384641  | 0.661672  |
| ENSG00000185973 | 0.6094381 | 0.462645  | 0.4754351 | -0.01278999 | 0.517875  | 0.661672  |
| ENSG00000168939 | 0.1462562 | 0.148196  | 0.1503039 | -0.00210801 | 0.509321  | 0.661672  |
| ENSG00000124333 | 0.8244133 | 0.547946  | 0.5753969 | -0.02745135 | 0.531678  | 0.661672  |

```
# select top HVGs
top_hvgs <- getTopHVGs(dec, prop = 0.05)
length(top_hvgs)
```

```
[1] 412
```

## Dimensionality reduction

Dimensionality reduction using PCA on the top highly variable genes revealed clear structure in the dataset. In the annotated PCA, spots separated primarily along a tumor–stromal axis on PC1, with additional variation associated with necrotic and vascular regions on PC2. Tumor spots were broadly dispersed, indicating multiple transcriptional subgroups, whereas lymphocytes formed a compact cluster positioned near stromal and CAF spots.

Application of the Leiden algorithm to the PCA neighbor graph identified six transcriptionally coherent clusters that matched the major groupings visible in PCA space. These clusters included distinct tumor-associated groups(4,5), stromal/CAF clusters(1,2,3), and a necrotic/blood vessel cluster(6), with some clusters showing intermediate positions between stromal and CAF populations.

UMAP projections provided a non-linear visualization of the same relationships, showing stromal, CAF, and immune spots forming a connected manifold, while tumor and necrotic/angiogenic clusters formed separate islands. Leiden labeling on the UMAP highlighted the same cluster structure and transitions observed in PCA, confirming consistent grouping of transcriptionally similar spots across both dimensionality reduction approaches.

## PCA

```
set.seed(12345)

speBreast <- runPCA(speBreast, subset_row = top_hvgs)
reducedDimNames(speBreast)
```

```
[1] "PCA"
```

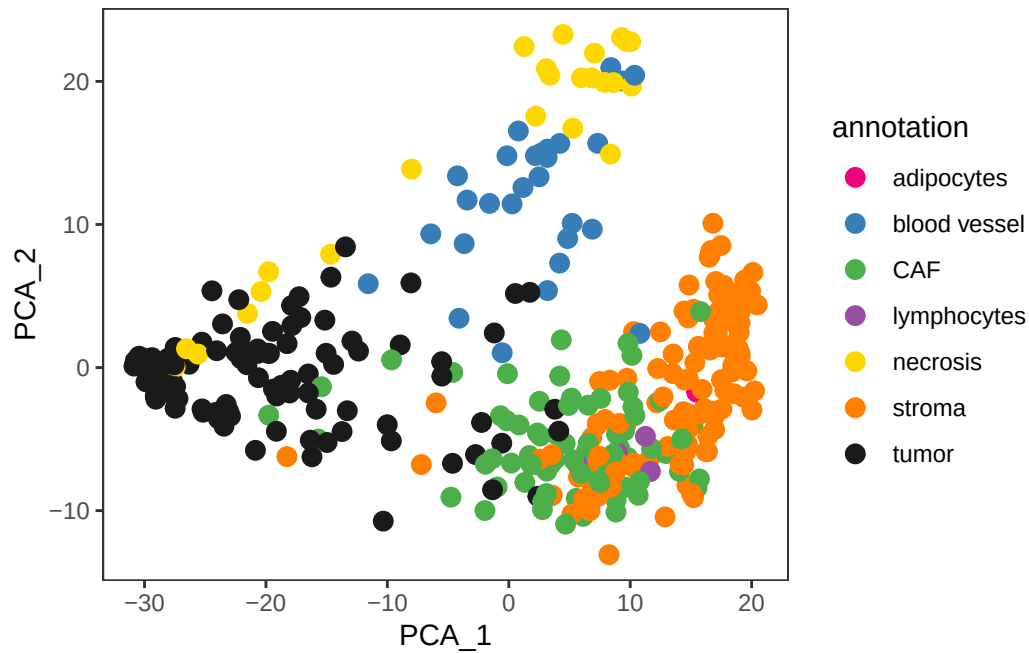
```
dim(reducedDim(speBreast, "PCA"))
```

```
[1] 348  50
```

```
colnames(reducedDim(speBreast, "PCA"))
```

```
[1] "PC1" "PC2" "PC3" "PC4" "PC5" "PC6" "PC7" "PC8" "PC9" "PC10"
[11] "PC11" "PC12" "PC13" "PC14" "PC15" "PC16" "PC17" "PC18" "PC19" "PC20"
[21] "PC21" "PC22" "PC23" "PC24" "PC25" "PC26" "PC27" "PC28" "PC29" "PC30"
[31] "PC31" "PC32" "PC33" "PC34" "PC35" "PC36" "PC37" "PC38" "PC39" "PC40"
[41] "PC41" "PC42" "PC43" "PC44" "PC45" "PC46" "PC47" "PC48" "PC49" "PC50"
```

```
# plot annotation in PCA reduced dimensions
plotDimRed(speBreast, plot_type = "PCA", annotate = "annotation",
           pal = "libd_layer_colors", point_size = 3)
```



## UMAP

```
speBreast <- runUMAP(speBreast, dimred = "PCA")
```

```
reducedDimNames(speBreast)
```

```
[1] "PCA" "UMAP"
```

```
dim(reducedDim(speBreast, "UMAP"))
```

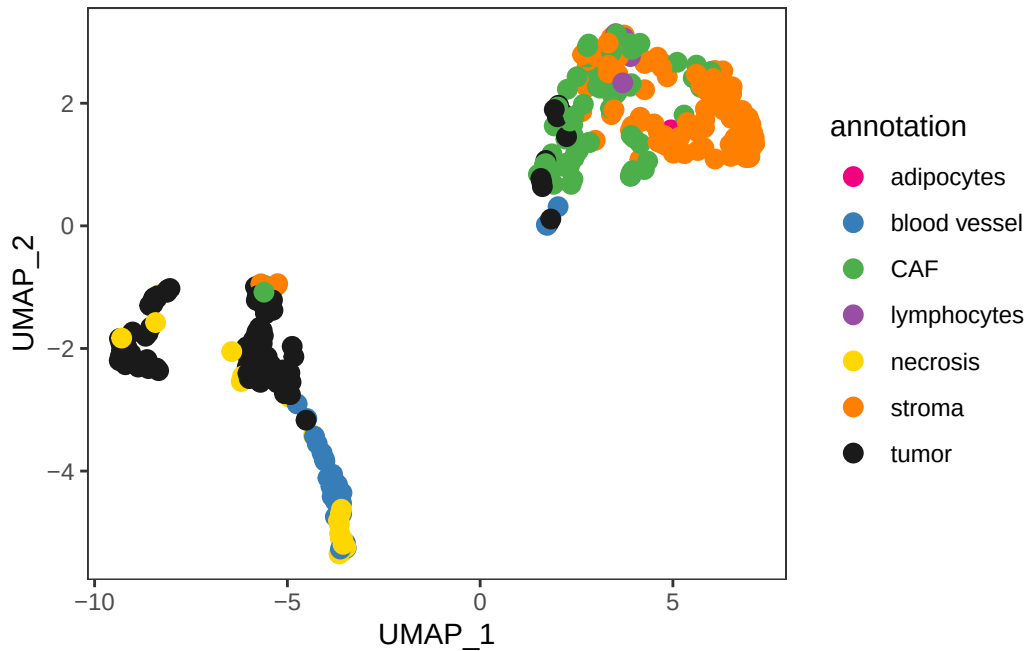
```
[1] 348  2
```

```
colnames(reducedDim(speBreast, "UMAP"))
```

```
[1] "UMAP1" "UMAP2"
```



```
# plot annotation in UMAP reduced dimensions
plotDimRed(speBreast, plot_type = "UMAP", annotate = "annotation",
           pal = "libd_layer_colors", point_size = 3)
```

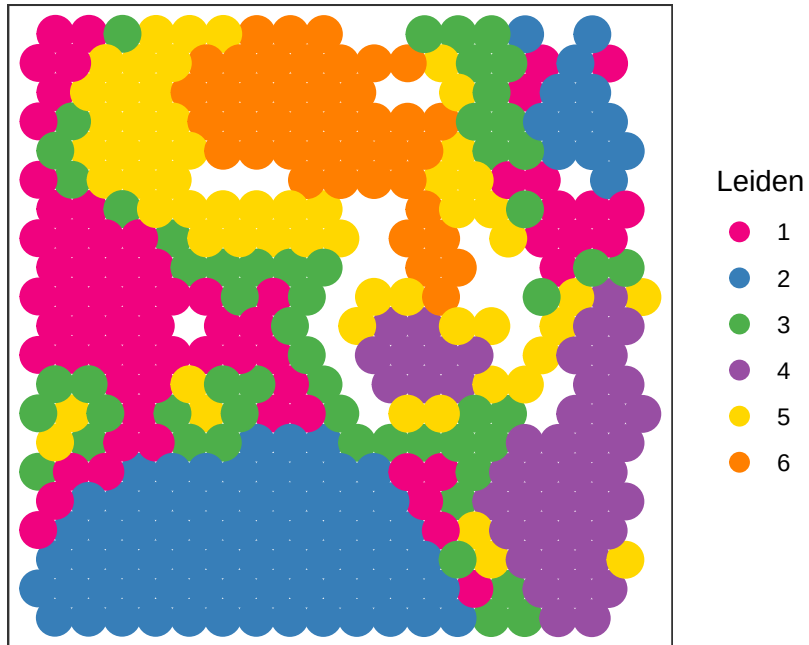


## Clustering

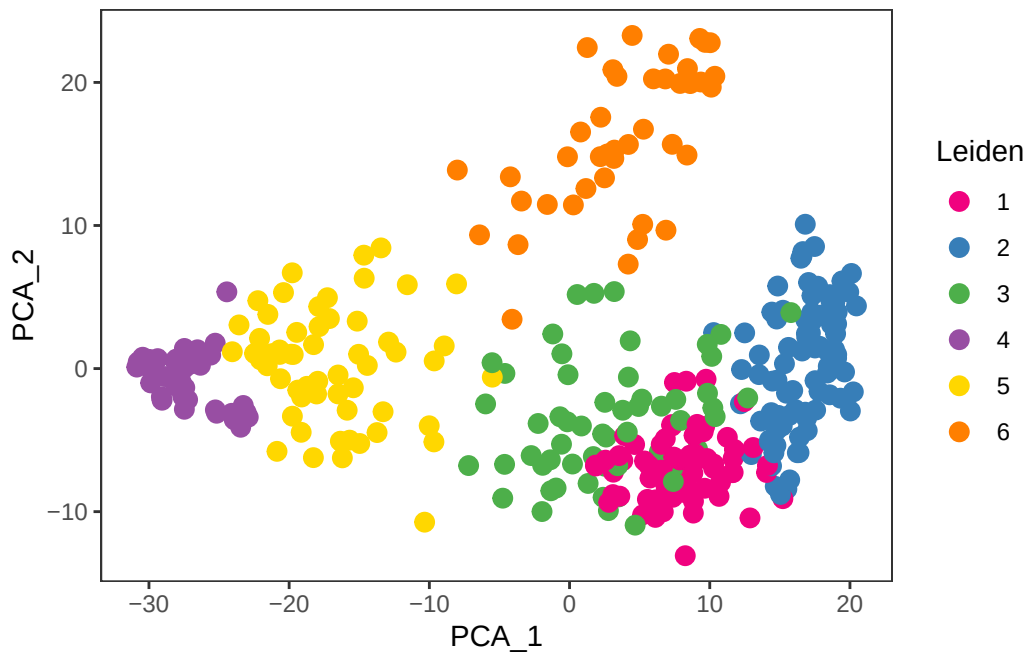
```
k <- 10
g <- buildSNNGraph(speBreast, k = k, use.dimred = "PCA")
k <- igraph::cluster_leiden(g, objective_function="modularity", resolution=1.2)
speBreast$Leiden <- factor(k$membership)
table(speBreast$Leiden)
```

```
 1  2  3  4  5  6
67 82 56 44 58 41
```

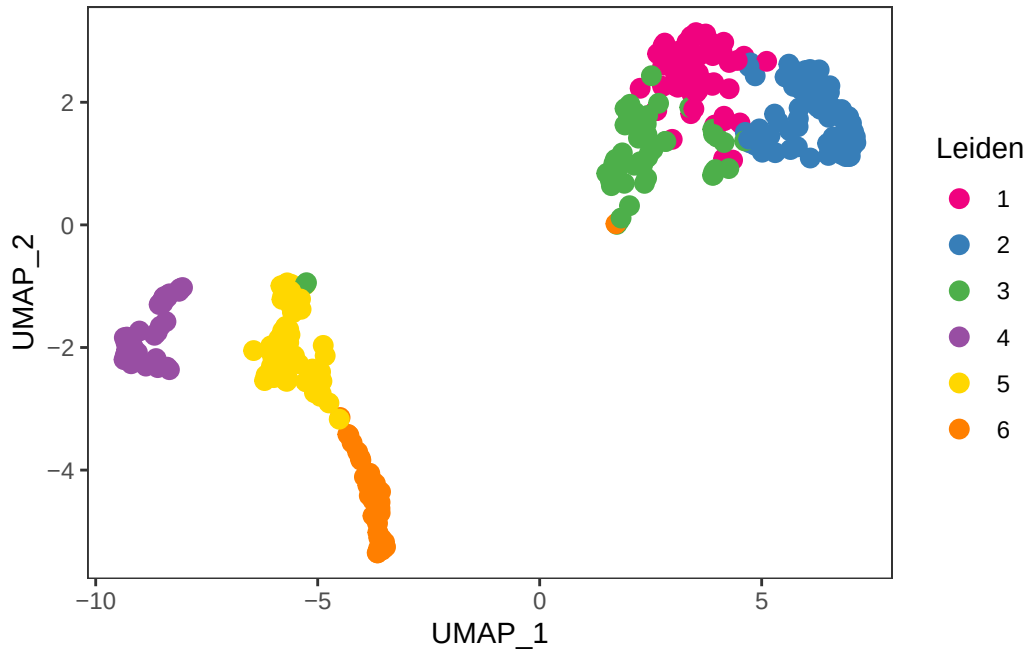
```
# plot clusters in spatial x-y coordinates
plotCoords(speBreast, annotate = "Leiden", pal = "libd_layer_colors", point_size = 6)
```



```
# plot clusters in PCA reduced dimensions
plotDimRed(speBreast, plot_type = "PCA", annotate = "Leiden",
  pal = "libd_layer_colors", point_size = 3)
```



```
# plot clusters in UMAP reduced dimensions
plotDimRed(speBreast, plot_type = "UMAP", annotate = "Leiden",
           pal = "libd_layer_colors", point_size = 3)
```



```
table(
  Leiden = speBreast$Leiden,
  Annotation = speBreast$annotation
)
```

|   | Annotation |            |              |     |             |          |        |       |
|---|------------|------------|--------------|-----|-------------|----------|--------|-------|
|   | Leiden     | adipocytes | blood vessel | CAF | lymphocytes | necrosis | stroma | tumor |
| 1 |            | 0          | 0            | 30  | 4           | 0        | 33     | 0     |
| 2 |            | 1          | 0            | 6   | 0           | 0        | 75     | 0     |
| 3 |            | 0          | 3            | 37  | 0           | 0        | 4      | 12    |
| 4 |            | 0          | 0            | 0   | 0           | 4        | 0      | 40    |
| 5 |            | 0          | 1            | 4   | 0           | 4        | 1      | 48    |
| 6 |            | 0          | 24           | 0   | 0           | 17       | 0      | 0     |

## Differential Expression

When we performed differential expression across Leiden clusters, the heatmap of average gene expression showed distinct expression signatures between groups. Clusters 1 and 3 were

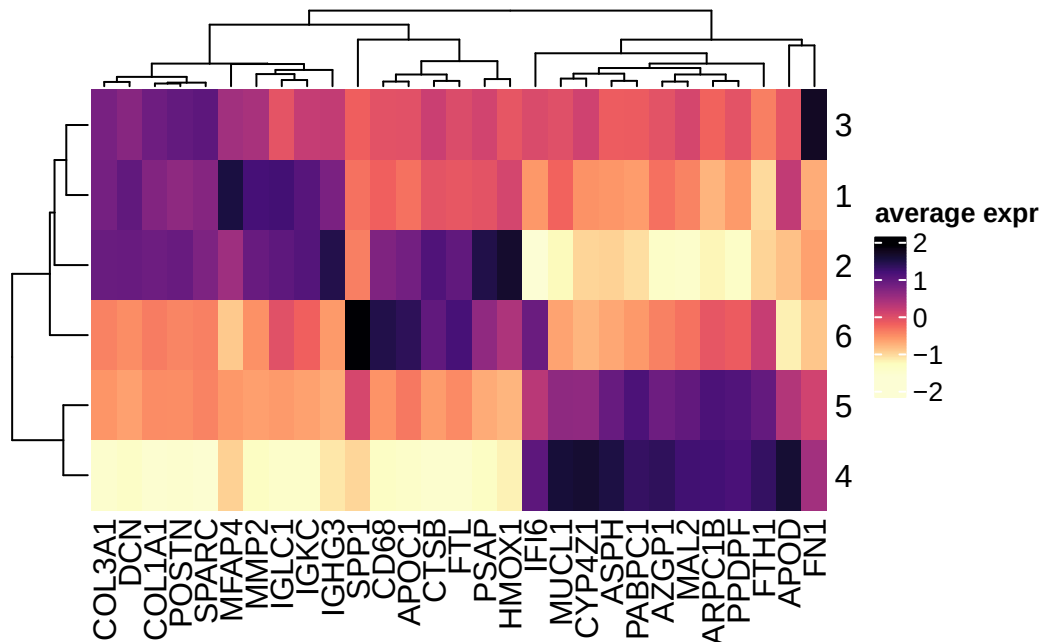
enriched for extracellular matrix and stromal genes such as **COL1A1**, **COL3A1**, **DCN**, **POSTN**, **SPARC**, and **MMP2**, indicating strong matrix-remodeling and fibroblast/CAF activity. Cluster 2 was characterized by high expression of **immunoglobulin genes** (**IGHG3**, **IGLC1**, **IGKC**) and **CD68**, consistent with immune and myeloid cell-associated expression. Clusters 5 and 6 showed elevated expression of **HMOX1**, **FTL**, **PSAP**, and **IFI6**, genes linked to hypoxia and stress responses, while cluster 4 showed comparatively low expression of stromal and immune markers and higher tumor-associated genes such as **MUCL1**, **MAL2**, and **AZGP1**.

```
mgs <- findMarkers(speBreast, groups = speBreast$Leiden, direction = "up")
top <- lapply(mgs, \(df) rownames(df)[df$Top <= 2])
top <- unique(unlist(top))
length(top)
```

[1] 29

```
# compute cluster-wise averages
pbs <- aggregateAcrossCells(speBreast, ids = speBreast$Leiden, subset.row = top,
                           use.assay.type = "logcounts", statistics = "mean")
# use gene symbols as feature names
mtx <- t(assay(pbs))
colnames(mtx) <- rowData(pbs)$symbol
```

```
library(ComplexHeatmap)
Heatmap(mat = scale(mtx), name = "average expr", col = c("#FCFDD4", rev(viridis::magma(10))))
```



These transcriptional profiles align with the dimensionality reduction findings. Clusters enriched in collagen and matrix genes correspond to the CAF/stromal regions seen in PCA/UMAP space, while immunoglobulin-rich clusters match the lymphocyte and immune populations that localized adjacent to stroma rather than tumor, supporting an immune-excluded architecture. The hypoxia/stress-associated clusters are consistent with the necrotic/angiogenic regions identified along PC2. Overall, the heatmap confirms the presence of distinct tumor, CAF/stromal, immune, and hypoxic/angiogenic compartments previously identified by PCA, UMAP, and Leiden clustering.

## RESULTS AND DISCUSSION

We proposed that breast tumors contain spatially localized COXIS-high tumor nests adjacent to LRRC15 CAF-rich stroma, forming organized tumor–stroma ecosystems potentially linked to therapy resistance.

Dimensionality reduction and clustering revealed distinct transcriptional regions, which we broadly grouped as CAF-rich (clusters 1 and 3) and tumor-rich (clusters 4 and 5) based on marker-expression patterns in the heatmap.

To compare these compartments, we performed gene-signature scoring for tumor and CAF groups we obtained after clustering.

We also carried out Gene Ontology pathway enrichment analysis separately in CAF-rich and tumor-rich groups to identify dominant biological processes. These analyses allowed us to

assess how tumor and CAF regions differ transcriptionally and functionally, and whether their spatial organization supported our hypothesis.

```
# 1) Defining CAF vs tumor groups

caf_clusters <- c("1","3")
tumor_clusters <- c("4","5")

group <- ifelse(speBreast$Leiden %in% caf_clusters,
               "CAF",
               ifelse(speBreast$Leiden %in% tumor_clusters,
                     "Tumor",
                     NA))

speBreast$group <- group

# keeping only CAF and tumor
spe_sub_sig <- speBreast[, !is.na(speBreast$group)]

spe_sub_GO <- speBreast[, !is.na(speBreast$group)]
```

## GENE-SIGNATURE SCORING

```
library(signifinder)
```

```
library(org.Hs.eg.db)
```

Loading required package: AnnotationDbi

```
symbols <- mapIds(
  org.Hs.eg.db,
  keys = rownames(spe_sub_sig),
  keytype = "ENSEMBL",
  column = "SYMBOL",
  multiVals = "first"
)
```

'select()' returned 1:many mapping between keys and columns

```
rownames(spe_sub_sig) <- symbols
```

```
spe_sig <- multipleSign(  
  dataset = spe_sub_sig,  
  inputType = "rnaseq",  
  tissue = "breast",  
  whichAssay = "logcounts" # <<< replace with the normalized assay you actually have  
)
```

EMTSigCheng is using 90% of signature genes

HRDSSig is using 84% of signature genes

LRRC15CAFSig is using 100% of signature genes

ICBResponseSig is using 95% of responder signature genes

ICBResponseSig is using 87% of nonresponder signature genes

Warning in cor(x, sup\_resp, method = "pearson"): the standard deviation is zero  
Warning in cor(x, sup\_resp, method = "pearson"): the standard deviation is zero  
Warning in cor(x, sup\_resp, method = "pearson"): the standard deviation is zero

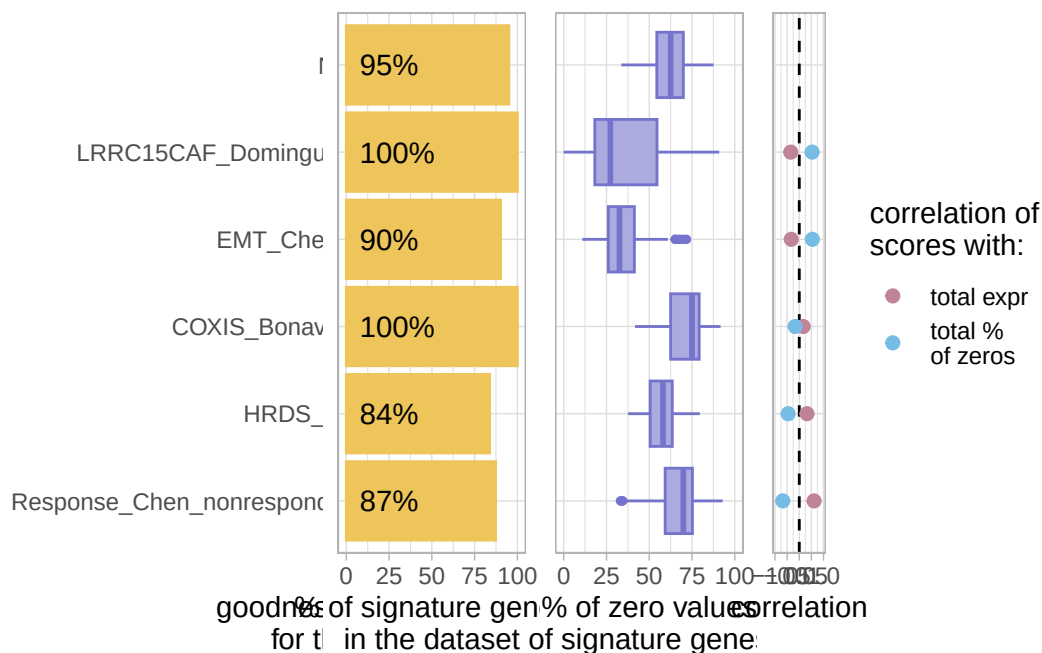
COXIISig is using 100% of signature genes

```
#checking signature goodness  
evaluationSignPlot(spe_sig, whichAssay = "logcounts")
```

Warning: Removed 2 rows containing missing values or values outside the scale range  
(`geom\_bar()`).

Warning: Removed 1 row containing missing values or values outside the scale range  
(`geom\_text()`).

Warning: Removed 1 row containing missing values or values outside the scale range  
(`geom\_point()`).  
Removed 1 row containing missing values or values outside the scale range  
(`geom\_point()`).



Several signatures showed high goodness-of-fit in the dataset, with LRRC15-CAF, EMT, and COXIS scoring highest and most signatures having >80% gene coverage. Zero-value proportions varied across signatures, and correlations with total expression or sparsity were generally low, indicating minimal technical bias.

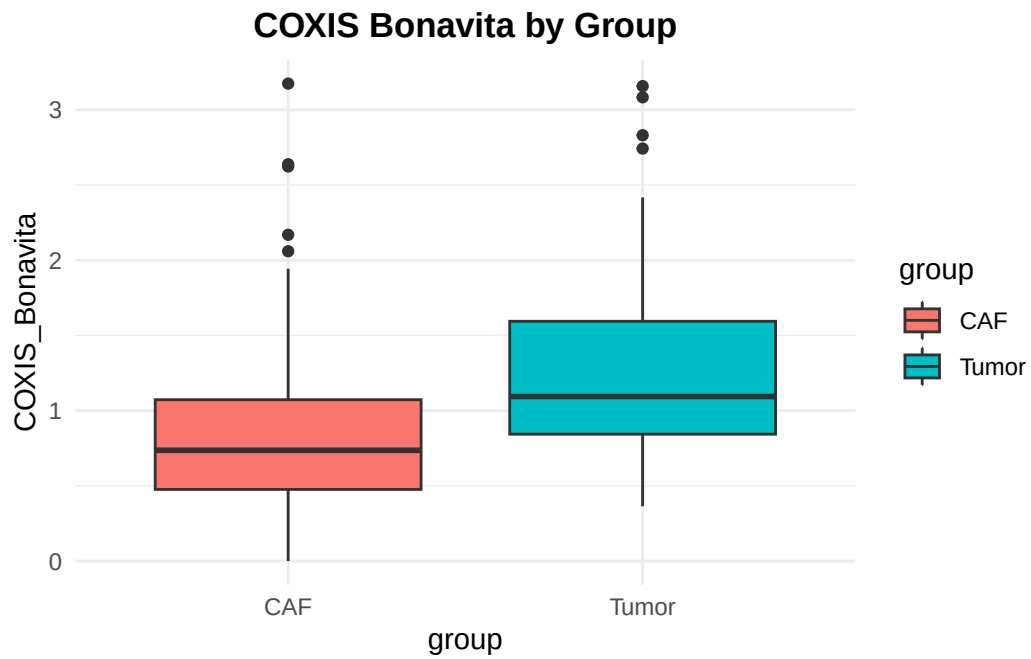
We then identified which biological programs were preferentially enriched in each compartment by comparing the two groups based on the gene-signatures scored.

Gene-signature scoring showed clear differences between the two groups. COXIS scores were significantly higher in tumor-rich clusters (4 and 5), while LRRC15-CAF and EMT signatures were significantly higher in CAF-rich clusters (1 and 3). Thus, tumor regions preferentially scored high for COXIS, whereas CAF-rich regions preferentially scored high for LRRC15-CAF and EMT programs.

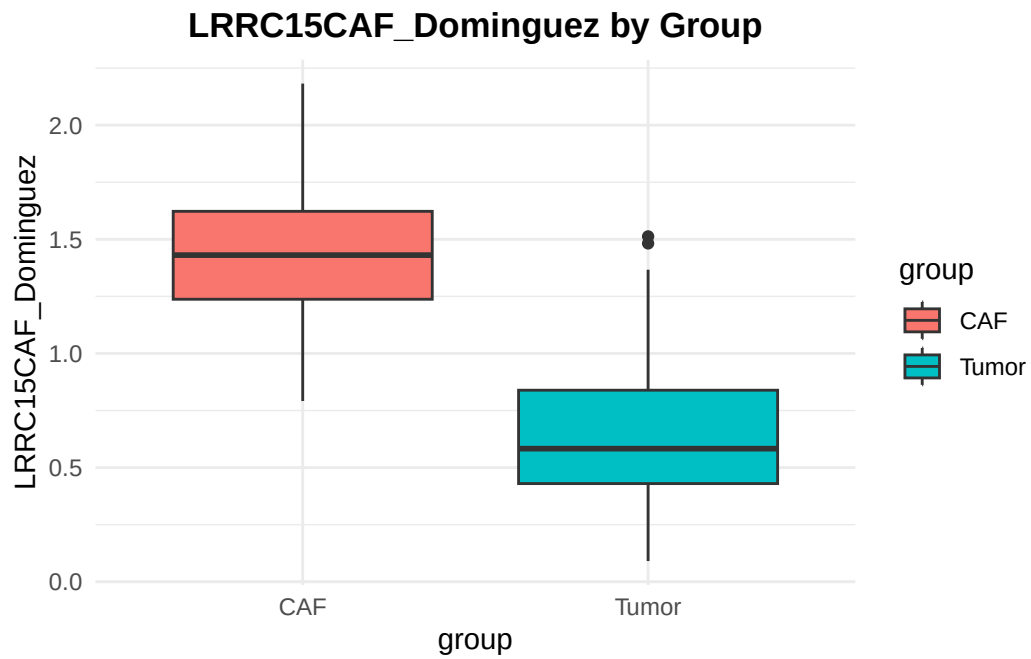
```
df <- as.data.frame(colData(spe_sig))

ggplot(df, aes(x = group, y =
COXIS_Bonavita, fill = group)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "COXIS Bonavita by Group") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```

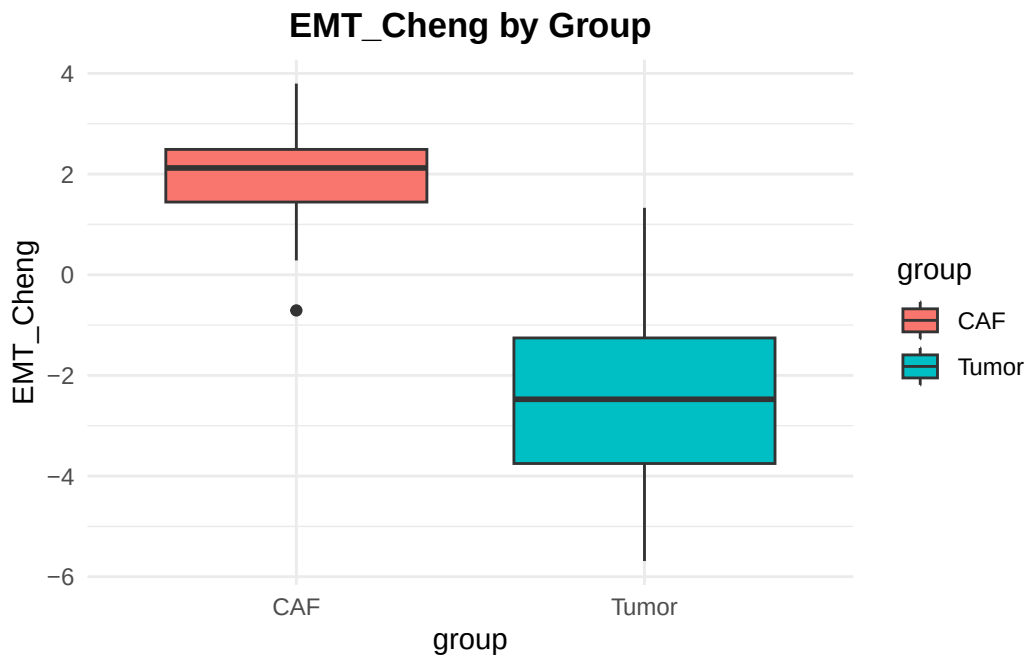




```
ggplot(df, aes(x = group, y = LRRC15CAF_Dominguez, fill = group)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs(title = "LRRC15CAF_Dominguez by Group") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



```
ggplot(df, aes(x = group, y = EMT_Cheng, fill = group)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs(title = "EMT_Cheng by Group") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



```
wilcox.test(COXIS_Bonavita ~ group, data = df)
```

Wilcoxon rank sum test with continuity correction

data: COXIS\_Bonavita by group

W = 3310.5, p-value = 1.104e-09

alternative hypothesis: true location shift is not equal to 0

```
wilcox.test(LRRC15CAF_Dominguez ~ group, data = df)
```

Wilcoxon rank sum test with continuity correction

data: LRRC15CAF\_Dominguez by group

W = 12101, p-value < 2.2e-16

alternative hypothesis: true location shift is not equal to 0

```
wilcox.test(EMT_Cheng ~ group, data = df)
```

Wilcoxon rank sum test with continuity correction

```
data: EMT_Cheng by group
W = 12454, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

The divergent signature profiles indicate distinct but coordinated tumor and stromal programs. Elevated COXIS scores in tumor-rich clusters are consistent with prior work showing activation of COXIS-related inflammatory and oxidative stress programs in tumor nests associated with therapeutic adaptation (*Bonavita et al., Cell 2020*). Higher LRRC15-CAF signatures in CAF-rich clusters agree with studies identifying LRRC15 CAFs as a specialized fibroblast population enriched in treated and resistant tumors (*Domínguez et al., Cancer Cell 2020; Elyada et al., Cancer Cell 2019*). Concurrent enrichment of EMT signatures within CAF-rich regions is consistent with the established role of CAFs in promoting EMT and invasion (*Nieto et al., Cell 2016; Cheng et al., Nat Commun 2021*). Taken together, these findings support a model in which COXIS-high tumor nests coexist with LRRC15 CAF-rich stroma as spatially organized tumor–stroma ecosystems linked to stress adaptation and therapy resistance (*Kieffer et al., Cancer Discovery 2020*).

## GENE ONTOLOGY PATHWAY ENRICHMENT

```
library(clusterProfiler)
```

clusterProfiler v4.16.0 Learn more at <https://yulab-smu.top/contribution-knowledge-mining/>

Please cite:

S Xu, E Hu, Y Cai, Z Xie, X Luo, L Zhan, W Tang, Q Wang, B Liu, R Wang, W Xie, T Wu, L Xie, G Yu. Using clusterProfiler to characterize multiomics data. Nature Protocols. 2024, 19(11):3292-3320

Attaching package: 'clusterProfiler'

The following object is masked from 'package:AnnotationDbi':

```
select
```

The following object is masked from 'package:igraph':

simplify

The following object is masked from 'package:IRanges':

slice

The following object is masked from 'package:S4Vectors':

rename

The following object is masked from 'package:stats':

filter

```
# Differential expression

de <- findMarkers(
  spe_sub_G0,
  groups = spe_sub_G0$group,
  direction = "up" # both up and down
)

# Inspection

caf_vs_tumor <- de$CAF[ de$CAF$FDR < 0.05, ] # genes up in CAF relative to tumor
tumor_vs_caf <- de$Tumor[ de$Tumor$FDR < 0.05, ] # genes up in tumor relative to CAF

head(caf_vs_tumor)
```

DataFrame with 6 rows and 5 columns

|                 | Top       | p.value     | FDR         | summary.logFC | logFC.Tumor |
|-----------------|-----------|-------------|-------------|---------------|-------------|
|                 | <integer> | <numeric>   | <numeric>   | <numeric>     | <numeric>   |
| ENSG00000132386 | 1         | 6.25708e-64 | 9.94750e-60 | 2.20036       | 2.20036     |
| ENSG00000163520 | 2         | 1.31411e-62 | 1.04458e-58 | 2.21299       | 2.21299     |
| ENSG00000011465 | 3         | 2.00439e-62 | 1.06219e-58 | 2.99505       | 2.99505     |
| ENSG00000159403 | 4         | 2.84860e-61 | 1.13218e-57 | 2.43602       | 2.43602     |
| ENSG00000087245 | 5         | 3.00082e-60 | 9.54140e-57 | 2.53610       | 2.53610     |
| ENSG00000038427 | 6         | 1.35812e-57 | 3.59857e-54 | 2.13361       | 2.13361     |

```
head(tumor_vs_caf)
```

DataFrame with 6 rows and 5 columns

|                 | Top       | p.value     | FDR         | summary.logFC | logFC.CAF |
|-----------------|-----------|-------------|-------------|---------------|-----------|
|                 | <integer> | <numeric>   | <numeric>   | <numeric>     | <numeric> |
| ENSG00000070756 | 1         | 2.16872e-69 | 3.44784e-65 | 1.07729       | 1.07729   |
| ENSG00000167996 | 2         | 1.50743e-67 | 1.19826e-63 | 1.41395       | 1.41395   |
| ENSG00000125534 | 3         | 1.01889e-60 | 5.39946e-57 | 1.36827       | 1.36827   |
| ENSG00000160862 | 4         | 1.27510e-59 | 5.06788e-56 | 1.70512       | 1.70512   |
| ENSG00000130429 | 5         | 4.29151e-59 | 1.36453e-55 | 1.10969       | 1.10969   |
| ENSG00000272398 | 6         | 8.59861e-56 | 2.27835e-52 | 1.53675       | 1.53675   |

```
# Ensemble to ENTREIDz
```

```
caf_genes <- rownames(caf_vs_tumor)
```

```
caf_entrez <- bitr(  
  caf_genes,  
  fromType = "ENSEMBL",  
  toType = "ENTREZID",  
  OrgDb = org.Hs.eg.db  
)
```

'select()' returned 1:many mapping between keys and columns

Warning in bitr(caf\_genes, fromType = "ENSEMBL", toType = "ENTREZID", OrgDb = org.Hs.eg.db): 0.32% of input gene IDs are fail to map...

```
tumor_genes <- rownames(tumor_vs_caf)
```

```
tumor_entrez <- bitr(  
  tumor_genes,  
  fromType = "ENSEMBL",  
  toType = "ENTREZID",  
  OrgDb = org.Hs.eg.db  
)
```

'select()' returned 1:many mapping between keys and columns

Warning in bitr(tumor\_genes, fromType = "ENSEMBL", toType = "ENTREZID", : 0.28% of input gene IDs are fail to map...

```
# Enrichment -----

# CAF

ego_caf <- enrichGO(
  gene = caf_entrez$ENTREZID,
  OrgDb = org.Hs.eg.db,
  ont = "BP",
  pAdjustMethod = "BH",
  readable = TRUE
)

# TUMOR

ego_tumor <- enrichGO(
  gene = tumor_entrez$ENTREZID,
  OrgDb = org.Hs.eg.db,
  ont = "BP",
  readable = TRUE
)

head(ego_caf)
```

| ID                      | Description                                   | GeneRatio      |          |              |
|-------------------------|-----------------------------------------------|----------------|----------|--------------|
| G0:0045785 G0:0045785   | positive regulation of cell adhesion          | 183/2316       |          |              |
| G0:0007159 G0:0007159   | leukocyte cell-cell adhesion                  | 162/2316       |          |              |
| G0:0030198 G0:0030198   | extracellular matrix organization             | 136/2316       |          |              |
| G0:0043062 G0:0043062   | extracellular structure organization          | 136/2316       |          |              |
| G0:0050867 G0:0050867   | positive regulation of cell activation        | 148/2316       |          |              |
| G0:0045229 G0:0045229   | external encapsulating structure organization | 136/2316       |          |              |
| BgRatio                 | RichFactor                                    | FoldEnrichment | zScore   | pvalue       |
| G0:0045785 499/18805    | 0.3667335                                     | 2.977730       | 16.78100 | 9.733445e-46 |
| G0:0007159 433/18805    | 0.3741339                                     | 3.037819       | 16.07787 | 8.302891e-42 |
| G0:0030198 339/18805    | 0.4011799                                     | 3.257422       | 15.71894 | 4.411996e-39 |
| G0:0043062 340/18805    | 0.4000000                                     | 3.247841       | 15.67573 | 6.493191e-39 |
| G0:0050867 392/18805    | 0.3775510                                     | 3.065564       | 15.48876 | 8.831100e-39 |
| G0:0045229 341/18805    | 0.3988270                                     | 3.238317       | 15.63267 | 9.537413e-39 |
| p.adjust                | qvalue                                        |                |          |              |
| G0:0045785 5.846881e-42 | 3.887231e-42                                  |                |          |              |

```
G0:0007159 2.493773e-38 1.657956e-38
G0:0030198 8.834286e-36 5.873373e-36
G0:0043062 9.548540e-36 6.348236e-36
G0:0050867 9.548540e-36 6.348236e-36
G0:0045229 9.548540e-36 6.348236e-36
```

```
G0:0045785 FBLN2/SFRP2/FBLN1/HLA-DRA/EMILIN1/HLA-DPA1/CD74/HLA-DPB1/CCN1/CCDC80/THY1/CXCL12/
G0:0007159
G0:0030198
G0:0043062
G0:0050867
G0:0045229
```

|            | Count |
|------------|-------|
| G0:0045785 | 183   |
| G0:0007159 | 162   |
| G0:0030198 | 136   |
| G0:0043062 | 136   |
| G0:0050867 | 148   |
| G0:0045229 | 136   |

```
head(ego_tumor)
```

| ID                    | Description                                         |              |                |          |          |              |
|-----------------------|-----------------------------------------------------|--------------|----------------|----------|----------|--------------|
| G0:1901293 G0:1901293 | nucleoside phosphate biosynthetic process           |              |                |          |          |              |
| G0:0048193 G0:0048193 | Golgi vesicle transport                             |              |                |          |          |              |
| G0:0072594 G0:0072594 | establishment of protein localization to organelle  |              |                |          |          |              |
| G0:0007005 G0:0007005 | mitochondrion organization                          |              |                |          |          |              |
| G0:0031647 G0:0031647 | regulation of protein stability                     |              |                |          |          |              |
| G0:0015980 G0:0015980 | energy derivation by oxidation of organic compounds |              |                |          |          |              |
| GeneRatio             | BgRatio                                             | RichFactor   | FoldEnrichment | zScore   | pvalue   |              |
| G0:1901293            | 102/3254                                            | 322/18805    | 0.3167702      | 1.830628 | 6.877063 | 1.632694e-10 |
| G0:0048193            | 94/3254                                             | 306/18805    | 0.3071895      | 1.775261 | 6.254452 | 4.891087e-09 |
| G0:0072594            | 129/3254                                            | 467/18805    | 0.2762313      | 1.596352 | 5.969525 | 1.312450e-08 |
| G0:0007005            | 124/3254                                            | 445/18805    | 0.2786517      | 1.610340 | 5.960325 | 1.444457e-08 |
| G0:0031647            | 110/3254                                            | 386/18805    | 0.2849741      | 1.646877 | 5.874043 | 2.577149e-08 |
| G0:0015980            | 102/3254                                            | 352/18805    | 0.2897727      | 1.674609 | 5.844460 | 3.278340e-08 |
|                       | p.adjust                                            | qvalue       |                |          |          |              |
| G0:1901293            | 1.003454e-06                                        | 8.551879e-07 |                |          |          |              |
| G0:0048193            | 1.503031e-05                                        | 1.280950e-05 |                |          |          |              |
| G0:0072594            | 2.219408e-05                                        | 1.891478e-05 |                |          |          |              |
| G0:0007005            | 2.219408e-05                                        | 1.891478e-05 |                |          |          |              |
| G0:0031647            | 3.167832e-05                                        | 2.699767e-05 |                |          |          |              |



GO:0015980 3.339785e-05 2.846313e-05

GO:1901293

GO:0048193

GO:0072594 HSPD1/CDH1/JUP/SEC63/XBP1/HSPA8/SUMO1/PKIA/USP36/CACNG4/SORT1/HSP90AA1/WBP2/VP

GO:0007005 HSPD1/TFRC/NDUFB9/HSP90AA1/POLDIP2/CXADR/CLU/FIS1/IFI6/ERBB4/NDUFB2/YME1L1/BIK/TI

GO:0031647

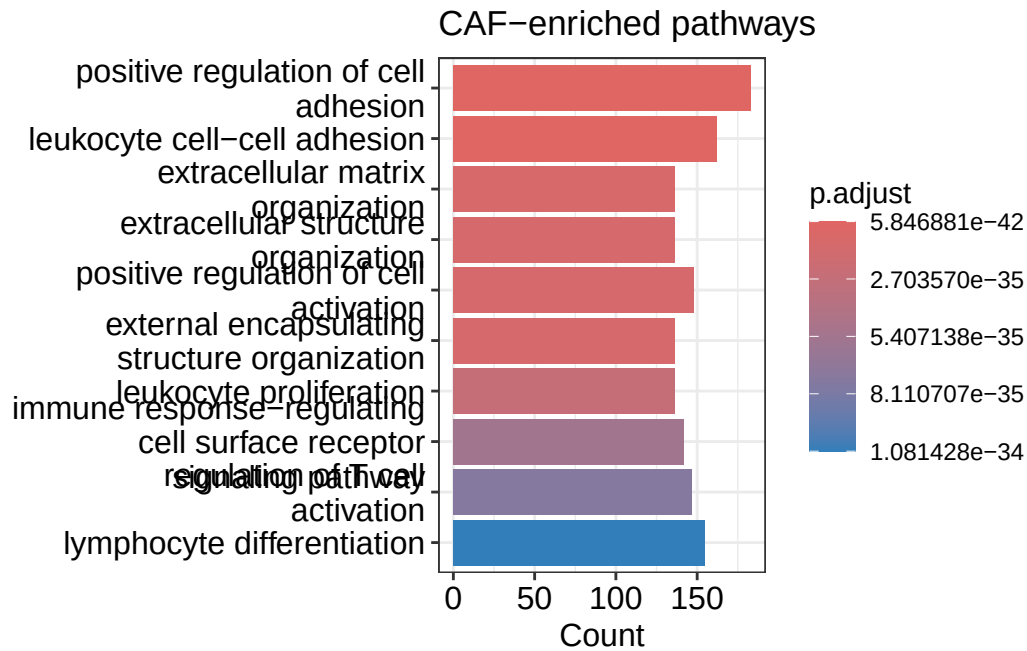
GO:0015980

|            | Count |
|------------|-------|
| GO:1901293 | 102   |
| GO:0048193 | 94    |
| GO:0072594 | 129   |
| GO:0007005 | 124   |
| GO:0031647 | 110   |
| GO:0015980 | 102   |

Gene Ontology enrichment analysis revealed distinct pathway programs in CAF-rich and tumor-rich clusters. CAF-rich clusters were enriched for pathways related to extracellular matrix organization, extracellular structure organization, positive regulation of cell adhesion, leukocyte cell-cell adhesion, and regulation of T-cell activation and lymphocyte differentiation. In contrast, tumor-rich clusters were enriched for pathways associated with nucleoside phosphate biosynthesis, mitochondrial organization, aerobic respiration, macromolecule catabolic processes, regulation of protein stability, and energy metabolism. These differences indicate that CAF-rich regions are dominated by adhesion, matrix, and immune-regulatory pathways, whereas tumor-rich regions are dominated by metabolic and biosynthetic programs.

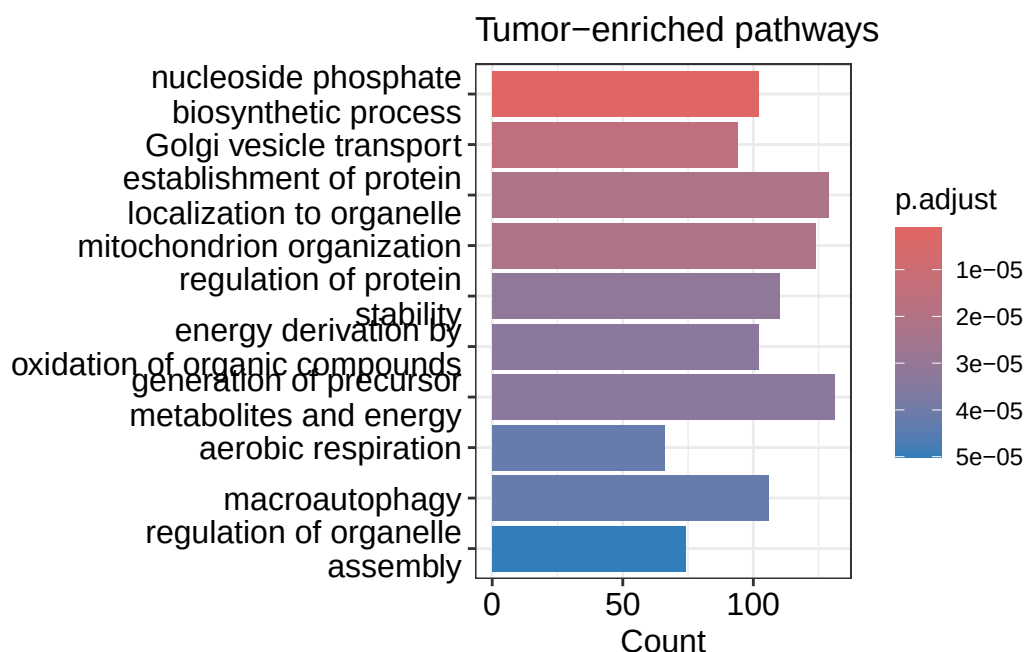
```
barplot(ego_caf, showCategory = 10, title = "CAF-enriched pathways")
```

```
Warning in fortify(object, showCategory = showCategory, by = x, ...): Arguments in `...` must
x Problematic argument:
* by = x
i Did you misspell an argument name?
```



```
barplot(ego_tumor, showCategory = 10, title = "Tumor-enriched pathways")
```

Warning in fortify(object, showCategory = showCategory, by = x, ...): Arguments in `...` must be the same length  
 x Problematic argument:  
 \* by = x  
 i Did you misspell an argument name?



The pathway enrichment analysis further supports the presence of specialized tumor–stroma ecosystems. CAF-rich clusters showed strong enrichment of extracellular matrix organization and cell adhesion pathways, consistent with matrix-remodeling and physical barrier functions of activated CAFs (*Kalluri, Nat Rev Cancer 2016; Elyada et al., Cancer Cell 2019*). Enrichment of leukocyte adhesion, regulation of T-cell activation, and lymphocyte differentiation pathways suggests that CAF regions actively participate in immune regulation and immune exclusion, in line with reports linking LRRC15 CAFs and TGF–driven stroma to immune checkpoint blockade resistance (*Dominguez et al., Cancer Cell 2020; Mariathasan et al., Nature 2018*). Conversely, tumor-rich clusters were enriched in mitochondrial organization, oxidative metabolism, nucleotide biosynthesis, and protein stability pathways, reflecting elevated metabolic and biosynthetic demands of proliferating tumor nests (*Pavlova & Thompson, Cell Metab 2016; Vander Heiden et al., Science 2009*). Together with the signature analysis, these pathway programs support a model in which metabolically active, COXIS-high tumor nests coexist with LRRC15 CAF-rich, matrix-remodeling stroma that modulates immune activation, forming a spatially organized microenvironment associated with treatment adaptation and immune evasion.

## CONCLUSION

Our spatial transcriptomic analysis supports the hypothesis that COXIS-high tumor nests are spatially adjacent to LRRC15 CAF-rich stroma, forming organized tumor–stroma ecosystems in breast tumors. Dimensionality reduction, clustering, signature scoring, and pathway enrichment

collectively showed that tumor-rich clusters were COXIS-high and metabolically active, while CAF-rich clusters were LRRC15-CAF-high, EMT-high, and enriched for extracellular matrix and immune-regulatory pathways.